

# Resources and Trade: The Heckscher-Ohlin Model

ECON 308 – International Economic Relations Chapter 5

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Fall 2026

# Opening Question: Why Does Canada Export Oil and China Export Electronics?

- Canada and China may use similar technology for many goods. The Ricardian model would predict little trade between them.
- Yet Canada is one of China's largest suppliers of oil, potash, and lumber, while China supplies Canada with electronics and consumer goods.
- The **H-O explanation**: each country exports the good that uses its *abundant factor* intensively.
- Canada is abundant in **land and natural resources**; China is abundant in **labour**. Their exports reflect their endowments.

| Country    | Abundant         | Exports               |
|------------|------------------|-----------------------|
| Canada     | Land/resources   | Oil, wheat, lumber    |
| China      | Labour           | Electronics, textiles |
| USA        | Capital/skills   | Aircraft, software    |
| Germany    | Skilled labour   | Machinery, cars       |
| Bangladesh | Unskilled labour | Garments              |

# Learning Objectives

1. Define factor abundance and factor intensity; state the H-O theorem.
2. Apply the Stolper-Samuelson theorem to predict distributional effects of trade.
3. Explain factor price equalisation and its limitations.
4. State and apply the Rybczynski theorem.
5. Evaluate the Leontief Paradox and modern evidence on H-O.
6. Discuss the debate on trade and wage inequality.

# Model Setup: Two Countries, Two Goods, Two Factors

## Assumptions:

- **2 countries:** Home (capital-abundant), Foreign (labour-abundant).
- **2 goods:** Cloth  $C$  (labour-intensive), Computers  $X$  (capital-intensive).
- **2 factors:** Labour  $L$  and Capital  $K$  — *both mobile* within each country (long-run).
- **Same technology** in both countries — the only difference is factor endowments.
- **Identical homothetic preferences:** both countries have the same demand structure at given prices.

### All variables:

$K, L$  = capital and labour endowments (Home)

$K^*, L^*$  = Foreign endowments

$K/L > K^*/L^*$ : Home is capital-abundant

$a_{Ki}, a_{Li}$  = capital and labour per unit of good  $i$

$r$  = rental rate of capital

$w$  = wage rate

$P_C, P_X$  = prices of cloth and computers

# Factor Intensity: The Key Concept

## Definition: Factor Intensity

Good  $X$  is **capital-intensive** relative to good  $C$  if at any given factor price ratio  $r/w$ :

$$\frac{K_X}{L_X} > \frac{K_C}{L_C}$$

i.e., computers always require more capital per worker than cloth does.

## Intuition:

- A semiconductor fab requires expensive equipment and relatively few workers.
- A garment factory requires many sewing machine operators and relatively little capital.
- This difference in factor intensity is what makes H-O work: capital-abundant.

| Industry         | $K/L$ ratio | Capital-int.?      |
|------------------|-------------|--------------------|
| Semiconductors   | Very high   | Yes                |
| Aircraft         | High        | Yes                |
| Automobiles      | Medium-high | Yes                |
| Apparel/textiles | Low         | No                 |
| Agriculture      | Medium-low  | No (in developing) |
| Call centres     | Very low    | No                 |

# The Heckscher-Ohlin Theorem

## H-O Theorem (Heckscher 1919, Ohlin 1933)

A country will **export** the good that uses its **abundant factor** intensively and **import** the good that uses its **scarce factor** intensively.

### Mechanism:

1. Home is capital-abundant:  $K/L > K^*/L^*$ .
2. Capital abundance  $\Rightarrow r/w$  is low in Home (capital is cheap relative to labour).
3. Low  $r/w \Rightarrow$  computers (capital-intensive) have lower relative cost in Home.
4. Home has comparative advantage in computers  $\Rightarrow$  Home exports computers.
5. Foreign exports cloth.

### Canada: H-O in Practice

Canada's factor endowments:

| Factor           | Status   | Key export            |
|------------------|----------|-----------------------|
| Land/resources   | Abundant | Oil, wheat, lumber    |
| Physical capital | Moderate | Aircraft (Bombardier) |
| Human capital    | Moderate | Financial services    |
| Unskilled labour | Scarce   | Import textiles       |

# Stolper-Samuelson Theorem: Trade and Factor Returns

## Stolper-Samuelson Theorem (1941)

An increase in the *relative price* of a good raises the **real return** to the factor used intensively in that good and lowers the real return to the other factor — in terms of *both* goods.

**Application:** Trade raises  $P_X/P_C$  in Home (capital-abundant exports computers):

- Real return to capital  $r/P$  rises in terms of *both cloth and computers*.
- Real wage  $w/P$  falls in terms of *both goods*.

**Magnification effect:**  $r$  rises by *more* than  $P_X$  rises;  $w$  falls in *absolute* terms.

## Long-Run Distributional Impact

In the long run (H-O model), trade harms the **scarce factor** absolutely:

- Home (capital-abundant): trade lowers *real wages* of workers.
- Foreign (labour-abundant): trade lowers *real returns to capital*.

This is harsher than the short-run ambiguity of Ch. 4.

# Stolper-Samuelson: Worked Example

**Suppose  $P_X$  rises by 10% while  $P_C$  is unchanged.**

With factor market clearing and cost minimisation:

- Firms in sector  $X$  raise wages and capital payments proportionally.
- But since  $X$  is capital-intensive, the benefit is concentrated in capital.
- Cloth sector now cannot afford as much capital (it's more expensive)  $\Rightarrow$  releases capital to  $X$  and hires more labour.
- Labour becomes more abundant in both sectors  $\Rightarrow w/P_C$  falls and  $w/P_X$  falls.
- Capital becomes scarcer in cloth  $\Rightarrow r/P_C$  and  $r/P_X$  both rise.

**In percentage terms (magnification):**

$$\hat{r} > \hat{P}_X > 0 > \hat{P}_C > \hat{w}$$

If  $P_X$  rises 10%:

$\hat{r} > 10\%$  (capital gains  $>$  price rise)

$\hat{w} < 0$  (labour loses in absolute terms)

**Both gains and losses exceed the price change.**



# Factor Price Equalisation

## Factor Price Equalisation Theorem (Samuelson, 1948)

Under free trade with identical technologies and no factor intensity reversals, **factor prices are equalised internationally** — even without factor mobility.

### Mechanism:

1. Trade equalises goods prices:  $P_X = P_X^*$  and  $P_C = P_C^*$ .
2. Same technology + same goods prices  $\Rightarrow$  same factor prices:  $w = w^*$ ,  $r = r^*$ .
3. Trade in goods *substitutes* for migration of factors.

**Strong prediction:** US wages should equal Chinese wages — clearly false!

### Why FPE Fails in Practice

- **Technology differs:** FPE requires identical technology.
- **Trade barriers:** tariffs and transport costs prevent full price equalisation.
- **Non-traded goods:** many goods/services are not traded.
- **Factor intensity reversals:** can invalidate FPE.
- **Incomplete specialisation:** if countries produce different goods, prices and factor

# Rybczynski Theorem: Growth and Trade

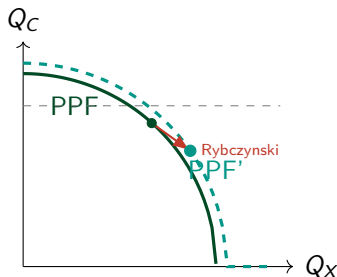
## Rybczynski Theorem (1955)

At **constant goods prices**, an increase in the endowment of one factor causes the output of the good using that factor intensively to **expand** and the output of the other good to **contract**.

**Example:** Home's capital stock grows ( $K \uparrow$ ):

1. Computers (capital-intensive): output **expands** — more capital available.
2. Cloth (labour-intensive): output **contracts** — capital expansion pulls labour out of cloth.

**Implication:** Growth *reinforces* comparative advantage. A capital-accumulating country becomes even more specialised in capital-intensive goods.



# Dutch Disease: Rybczynski in Canada

**Dutch Disease:** a natural resource boom expands the resource sector, which “crowds out” manufacturing by raising factor costs.

**Mechanism** (Rybczynski):

1. Oil sands investment ( $K$  in resources  $\uparrow$ ).
2. Resource output expands; manufacturing contracts.
3. CAD appreciates (higher demand for Canadian exports)  $\Rightarrow$  further hurts manufacturing competitiveness.

**Named after:** discovery of North Sea gas in Netherlands (1959), which caused manufacturing decline.

## Alberta Oil Sands 2000s

Investment in Alberta oil sands pulled capital and workers from Ontario/Quebec manufacturing.

Ontario manufacturing employment fell from 1.0M (2002) to 0.7M (2015) — partly Dutch Disease.

The CAD rose to parity with USD (2007–2014), making Ontario exports expensive.

# The Leontief Paradox

**Leontief (1953):** tested H-O with 1947 US data.

**H-O prediction:** the US is capital-abundant  $\Rightarrow$  US exports should be capital-intensive.

**Finding:** US *imports* were **30% more capital-intensive** than US exports — the opposite!

## Proposed explanations:

1. US exports are *human-capital* intensive (skill, not physical capital).
2. US has comparative advantage in *technology*, not capital per se.
3. Tariff structure: US tariffs protected labour-intensive industries.

## Resolution: Broaden “Capital”

When Trebler (1995) allows for *human capital* differences across countries and uses factor content of trade, H-O performs much better. The paradox largely disappears when we define factors more carefully.

## Key empirical papers:

Leontief (1953): original paradox  
Baldwin (1971): confirmed with 1962 data  
Trebler (1995): “Missing trade” puzzle  
Davis & Weinstein (2001): H-O with tech

# Trade, Technology, and Wage Inequality

## US wage inequality has risen since 1980:

- College–high school premium doubled.
- Top 1% income share surged.

## Two competing explanations:

### 1. Trade (Stolper-Samuelson):

- Trade with labour-abundant countries lowers unskilled wages.
- H-O predicts this in a capital/skill-abundant country.

### 2. Technology (SBTC):

- Skill-biased technical change raises relative demand for skilled workers.
- Computers and automation displace routine tasks.

| Factor           | Trade   | SBTC     |
|------------------|---------|----------|
| Unskilled labour | Lose    | Lose     |
| Skilled labour   | Gain    | Gain     |
| Capital          | Gain    | Gain     |
| Verdict          | Partial | Dominant |

### Consensus

Both trade and technology contribute. SBTC accounts for more of the rise in inequality (Autor et al.), but the China shock (Autor, Dorn, Hanson) shows trade effects are larger than previously thought.

# Key Takeaways

1. **H-O Theorem:** countries export goods using their abundant factor intensively — factor endowments drive comparative advantage.
2. **Stolper-Samuelson:** trade raises the real return to the abundant factor and lowers the real return to the scarce factor — in terms of *both* goods (magnification).
3. **FPE:** goods trade can equalise factor prices internationally — but strong assumptions limit its empirical validity.
4. **Rybczynski:** factor accumulation biases production toward the good using that factor intensively (Dutch Disease).
5. **Leontief Paradox:** H-O is empirically supported once human capital and technology differences are accounted for.
6. Trade *and* skill-biased technical change both contribute to rising wage inequality in rich countries.

Next: Chapter 6 — The Standard Trade Model

A unifying framework for terms of trade and welfare analysis.

# Discussion Questions

## Discussion

1. Is Canada labour-abundant or capital-abundant relative to Mexico? To China? How does your answer predict different trade patterns with each country?
2. The Rybczynski theorem says capital accumulation hurts the labour-intensive sector. Does this mean Canada's oil sands development harmed Ontario manufacturers?
3. FPE predicts wages should equalise with free trade. Why does this not alarm labour unions in Canada?

## Further reading:

- Heckscher, E. (1919); Ohlin, B. (1933): original H-O papers.
- Leontief, W. (1953), "Domestic Production and Foreign Trade," *Proceedings of the APS*.
- Trefler, D. (1995), "The Case of the Missing Trade," *AER*.
- Autor, D., Dorn, D., & Hanson, G. (2016), "The China Shock," *Annual Review of Economics*.